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STRATEGIES FOR TESTING AI-BASED AGENTS

ENSURING QUALITY AND RELIABILITY IN INTELLIGENT SYSTEMS

Orlando
30th April 2026

AGENDA

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- Introduction
- Why ?
  - 2026 View
  - Recipe of Failure
  - Traditional testing
- What ?
  - Typical agentic system and evaluation
  - MCP testing
  - A2A & AG-UI testing
  - Context based 4P framework
  - 3 Level evaluation model
  - E-2-E evaluation
- How ?
  - White boarding – 2 use cases

INTRODUCTION

Rajeev Sawant brings over 25 years of experience in the IT industry, specializing in quality engineering transformation, AI-driven testing, delivery management, and program leadership. A pioneer in intelligent script-less automation and digital testing, Rajeev combines his practical expertise with a Master's degree in AI/ML to seamlessly integrate advanced technologies into modern testing frameworks. He has a proven track record in managing global programs, establishing Testing Centers of Excellence (TCoE), driving process improvement, and fostering competency development. Rajeev is adept at transitioning traditional testing practices into innovative, AI-powered Quality Engineering solutions, with a focus on KPI-driven optimization and operational excellence.

Rajeev Sawant

Senior Manager,
Technology Consulting

EPAM India

2026 : AI IS NO LONGER AN EXPERIMENT

IT IS A DIGITAL WORKFORCE

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We're shifting from task-based automation to autonomous, role-based AI workforces, requiring enterprise-level quality, safety, and reliability strategies.

2026 Market and Regulatory Pressures

Financial Scrutiny

Forrester predicts a significant market correction. Fewer than 1/3 of decision-makers can link AI initiatives to financial growth. CFOs are deferring 25% of spend pending clear ROI.

Regulatory Mandates

The EU AI Act and ISO/IEC 42001 are no longer guidelines but strict, auditable requirements for market access and legal defense.

Legal Liability

Gartner predicts a catastrophic "black box" failures are projected to drive over 2,000 legal claims by the end of 2026.

Strategic Shift & Testing Implications

2025	2026
Operational Focus Task-based automation & POCs	**Operational Focus** Role-based autonomous agents in production
Financial Oversight Aggressive, unvetted AI spending	**Financial Oversight** CXO-driven ROI modeling
Regulatory Landscape Emerging guidelines	**Regulatory Landscape** Strict compliance mandates (ISO/IEC 42001)
Talent Ecosystem Specialised team at few locations	**Talent Ecosystem** Force Multiplier with special talent

95% OF GENERATIVE AI PILOTS AT COMPANIES ARE FAILING
(MIT REPORT – AUG 2025)

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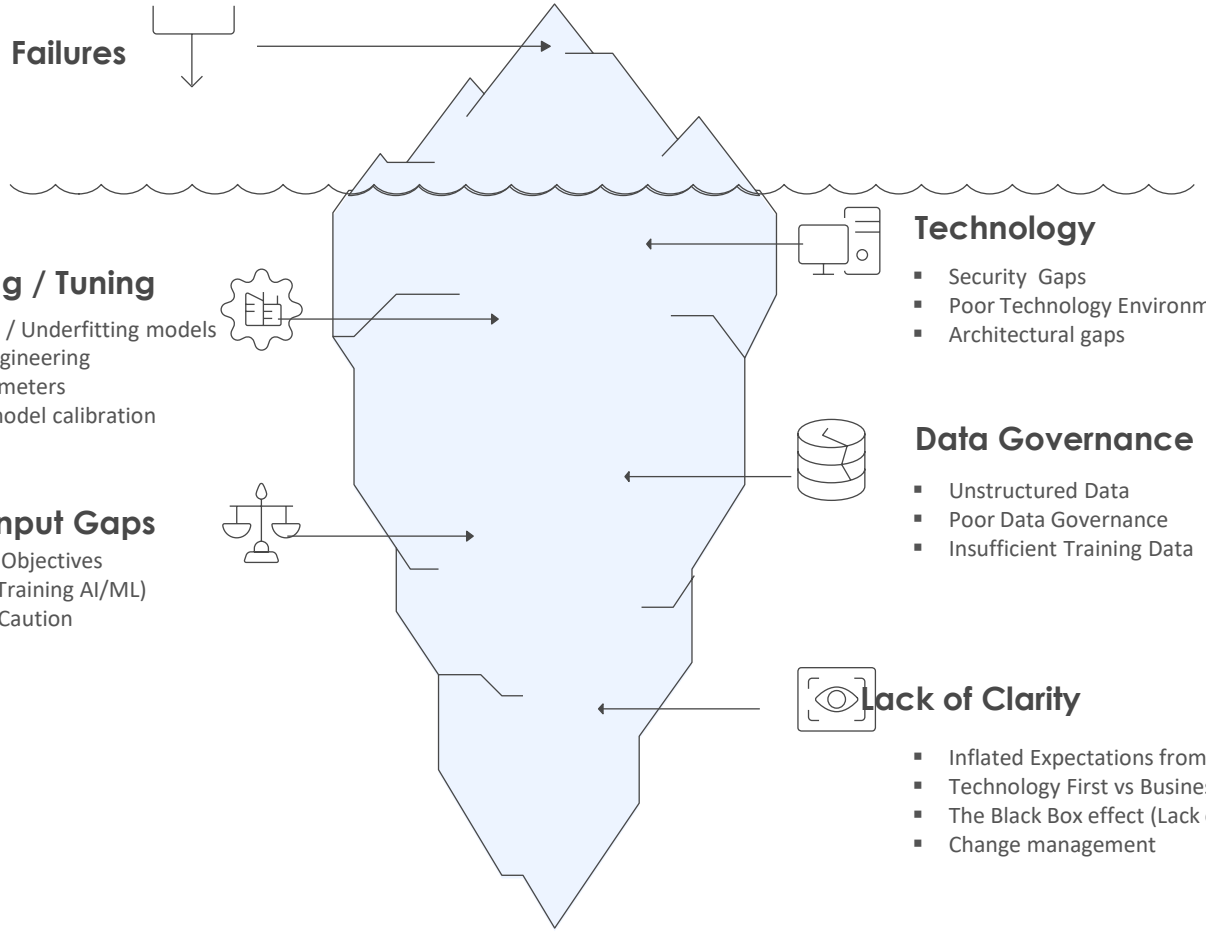
Artificial Intelligence : A double-edged sword with an even sharper edge

RECIPIE OF FAILURE

AI Failures



Building / Tuning

- Overfitting / Underfitting models
- Feature engineering
- Hyperparameters
- Dynamic model calibration

Input Gaps

- Ethics & Objectives
- Biases (Training AI/ML)
- Trust vs Caution

Technology

- Security Gaps
- Poor Technology Environment
- Architectural gaps

Data Governance

- Unstructured Data
- Poor Data Governance
- Insufficient Training Data

Lack of Clarity

- Inflated Expectations from AI Solutions
- Technology First vs Business First
- The Black Box effect (Lack of interpretability)
- Change management

IMPACT

INDIVIDUAL
- Physical Safety
- Privacy & Reputation
- Digital Safety
- Socio Economical

ORGANIZATIONAL
- Financial Performance
- Non-Financial
- Legal & Compliance
- Reputation

SOCIO ECONOMICAL
- Security
- Economic Stability
- Political Stability
- Infrastructure Integrity

TRADITIONAL TESTING APPROACHES ARE INADEQUATE TO TEST APPLICATIONS DEVELOPED USING AI

When we use outdated testing metrics to evaluate AI systems, we can end up with a "false sense of quality."

🔧 Traditional Software Testing:

- ✅ Deterministic Behavior
- 🧱 Component-based Validation.
- 📜 Specification-driven Testing
- 📉 Coverage Metrics
- ⚙️ Static Input/Output

🧠 Testing LLM-based Agentic Systems:

- 🔁 Non-deterministic Outputs
- 🔍 Goal-oriented Evaluation
- 🧠 Reasoning & Adaptability
- 📊 Qualitative Metrics
- 🧪 Prompt & Context Testing
- 🚦 Human-in-the-loop Testing

TYPICAL AGENTIC AI EVALUATION

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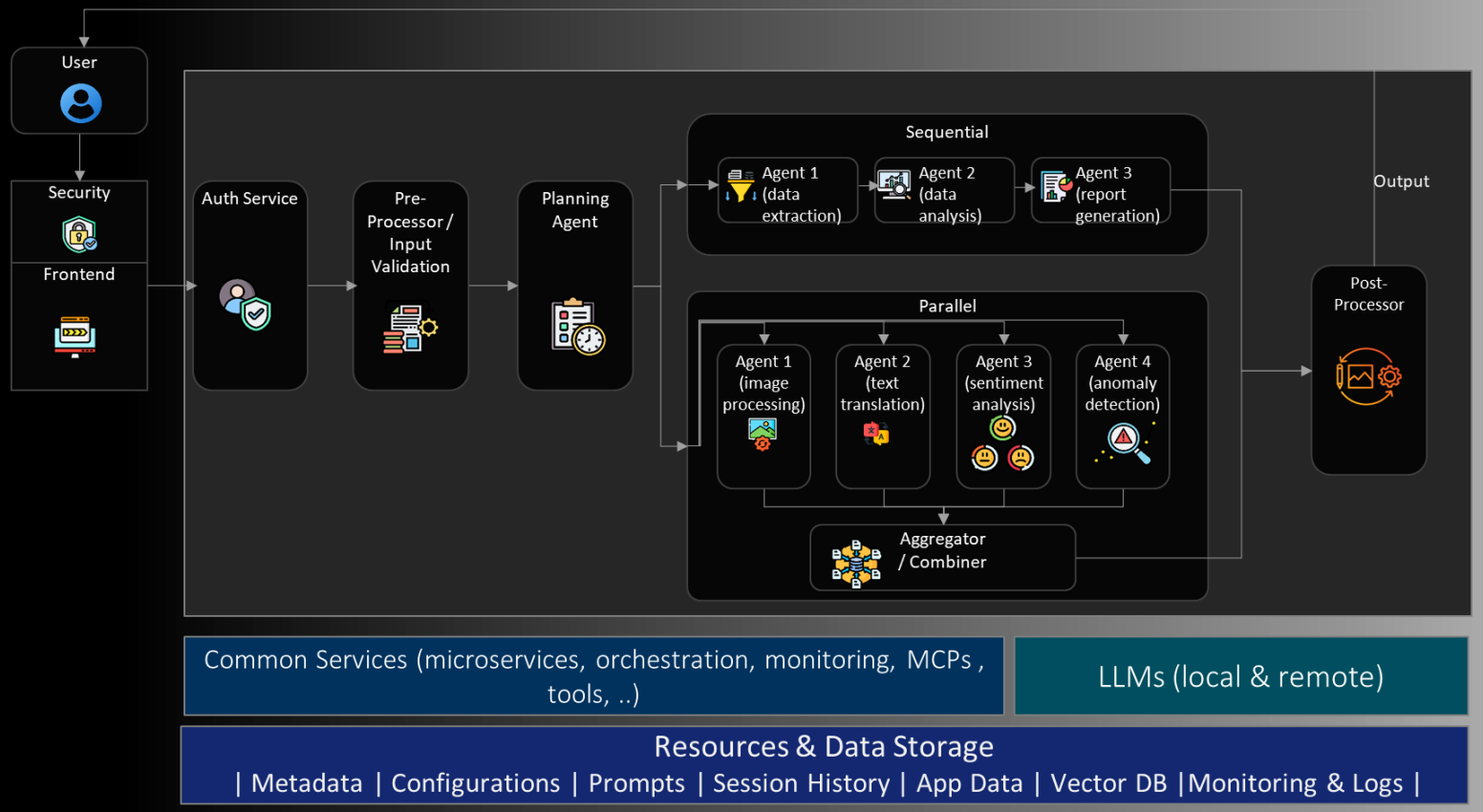
Sample Agentic Ecosystem



MCP TESTING : KEY TENETS

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The "Clarity" Tenet
Discovery & Tool Metadata

Agents "discover" tools via descriptions, the text is as important as the code.

- Description Quality
- Tools & Resources inventory

The "Reliability" Tenet
Functional Execution

Functional testing applied to the MCP interface

- Input Boundary Testing graceful handling of "hallucinated" parameters
- Error Mapping with specific error codes.
- Statefulness test for sequential operations

The "Safety" Tenet
Security & Isolation

MCP servers often have direct access to your local files or internal databases

- Least Privilege with limited access to outside directories or data
- Auth Persistence and handling of OAuth tokens or invalid API keys.
- Injection Guardrails through sanitized inputs prevent prompt injection

The "Performance" Tenet
The "Hit Rate" & "Success Rate"

Performance metrics for Agentic systems

- Tool Hit Rate : Correctly choose the tool for a given prompts
- Tool Success Rate: Successfully return / update data vs Timeout/ failure

The "Foundational" Tenet **Protocol Compliance**

Ensure server uses correct MCP protocol before AI testing.

- Validate all messages follow JSON-RPC Integrity
- Lifecycle Validation ensuring server advertises proper capabilities and reject invalid versions.
- Schema Strictness for input validation

A2A & AG-UI TESTING

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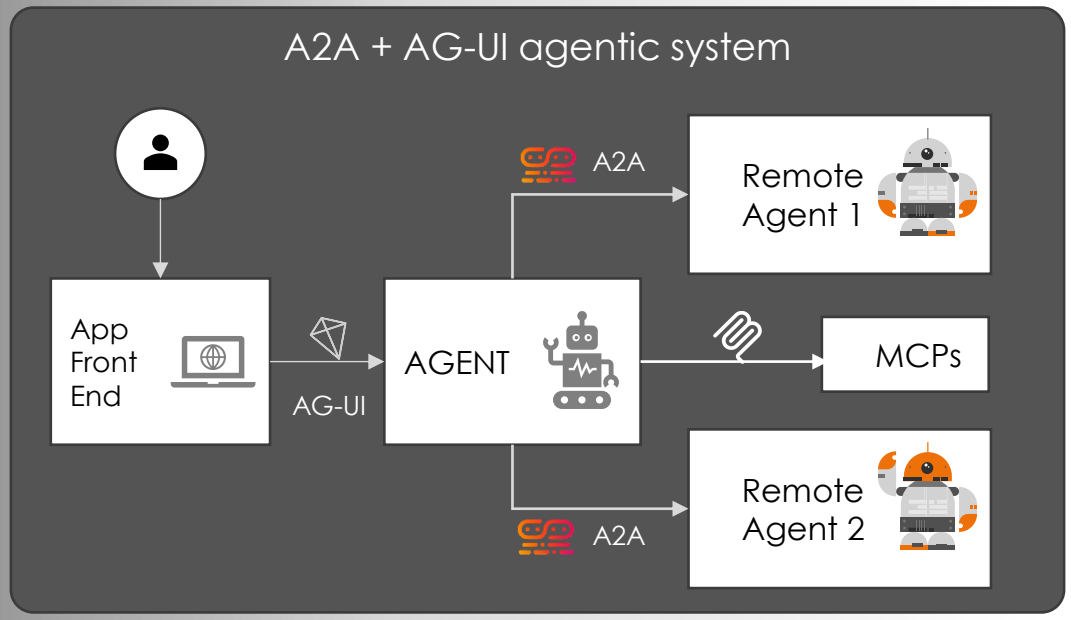
Key Focus Areas

A2A

- **Hand-off Integrity:**
  - Verify accurate context transfer between agents.
  - Assess for information loss during multi-agent task progression.
- **Protocol Consistency:**
  - Ensure unified communication formats are consistently followed.
  - Confirm no agent infers instructions not explicitly provided.
- **State Synchronization:**
  - Validate consistent shared state across all agents.
  - Test for race conditions in multi-step workflows.
- **Robustness Testing :** Inject conflicting instructions to evaluate system self-correction.
- **Deadlock Detection :** Identify infinite clarification loops between agents.

AG-UI

- **Semantic Interaction:** Assess if agents interpret affordances over technical identifiers.
- **UI Change Impact:** Test if functional recognition persists after UI modifications.
- **Action Validation:** Verify precise action selection amidst similar interface elements.
- **Latency Handling:** Evaluate agent response to incomplete or cached page states.
- **Visual Regression:** Confirm navigation capability despite CSS or style changes.
- **Negative Path Testing:** Test agent resilience against pop-ups and captchas.
- **State Persistence:** Ensure agents recognize cross-page data dependencies.



THE CONTEXT-BASED 4P FRAMEWORK

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1

Purpose (Reasoning Control)

- Define tasks and sub-goals
- Version prompts and roles
- Set success criteria & metrics

- Analyze new feature specifications.
- Use RAG to find related test cases.
- Generate impact reports.
- Maintain versioned prompts and role files in the repository.

2

Protection (Failure Prevention)

- Detect adversarial inputs
- Enforce policy at all stages
- Escalate on policy risk

- Check file size, validate schema, and remove profanity and PII
- Filter prompt injections in RAG retrieval queries
- Automatically refuse or escalate content containing banned material

3

Persistence (State & Memory Governance)

- Manage and recover memory
- Prune irrelevant tokens
- Track token and latency budgets

- The Vector DB holds past requirements and versioned test case embeddings.
- The cache retains data from the last three major releases; older data is archived.
- Outdated tests are removed each night.

4

Principles (Safety-Driven Behavior)

- Prevent unsafe actions
- Gate high-risk outputs
- Log policies and decisions [Audit]

- Destructive test cases require manual review before generation.
- All policy checks and denials are recorded for auditing.
- Adheres to the organization's Responsible AI charter.

THREE-LEVEL EVALUATION MODEL

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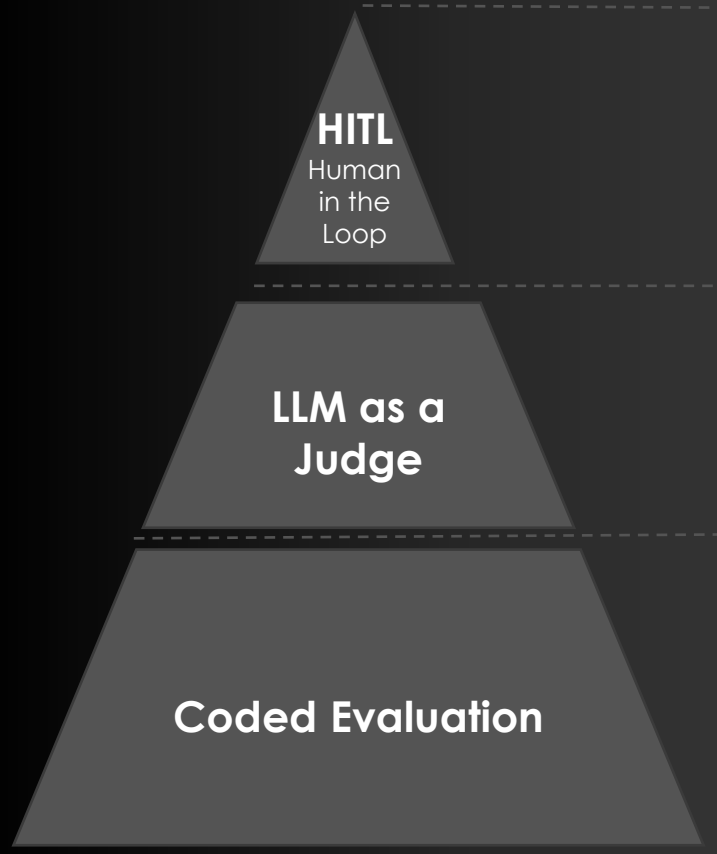
Assessing agentic behaviors requires combining **deterministic, semantic, and human evaluations**, as no single evaluator suffices.



HITL

- ✓ What: Human review of high-risk/ambiguous cases
- ✓ Role: Safety, ethics, gold standard
- ✓ When: [Selective]
- ✓ Trade-off: ✅ Highest accuracy, highest cost

LLM as a Judge

- ✓ What: LLM scores with explicit rubrics
- ✓ Role: Scales qualitative judgment, cost-efficient
- ✓ When: [Often]
- ✓ Trade-off: ✅ High nuance, scalable, lower cost

Coded Evaluation

- ✓ What: Deterministic, objective tests
- ✓ Role: Regression defence, CI/CD, structure
- ✓ When: [Always]
- ✓ Trade-off: ✅ Fastest, cheapest, widest coverage

Golden Set : Ground Truth

Serves as a regression benchmark for non-deterministic AI systems.

Relies on curated, high-fidelity queries over exact text matching.

1. **Input: Precise user prompt, including complex edge cases.**
2. **Context: Specific data or environment available to the agent.**
3. **Reference: Human-verified, exemplary response for quality and accuracy.**
4. **Evaluation Rubric: Clear grading criteria for agent performance.**

E-2-E EVALUATION APPROACH

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Evaluation Dataset Factory:
- Golden Dataset
- Synthetic Test Data
- Ground Truths/Reference Answers

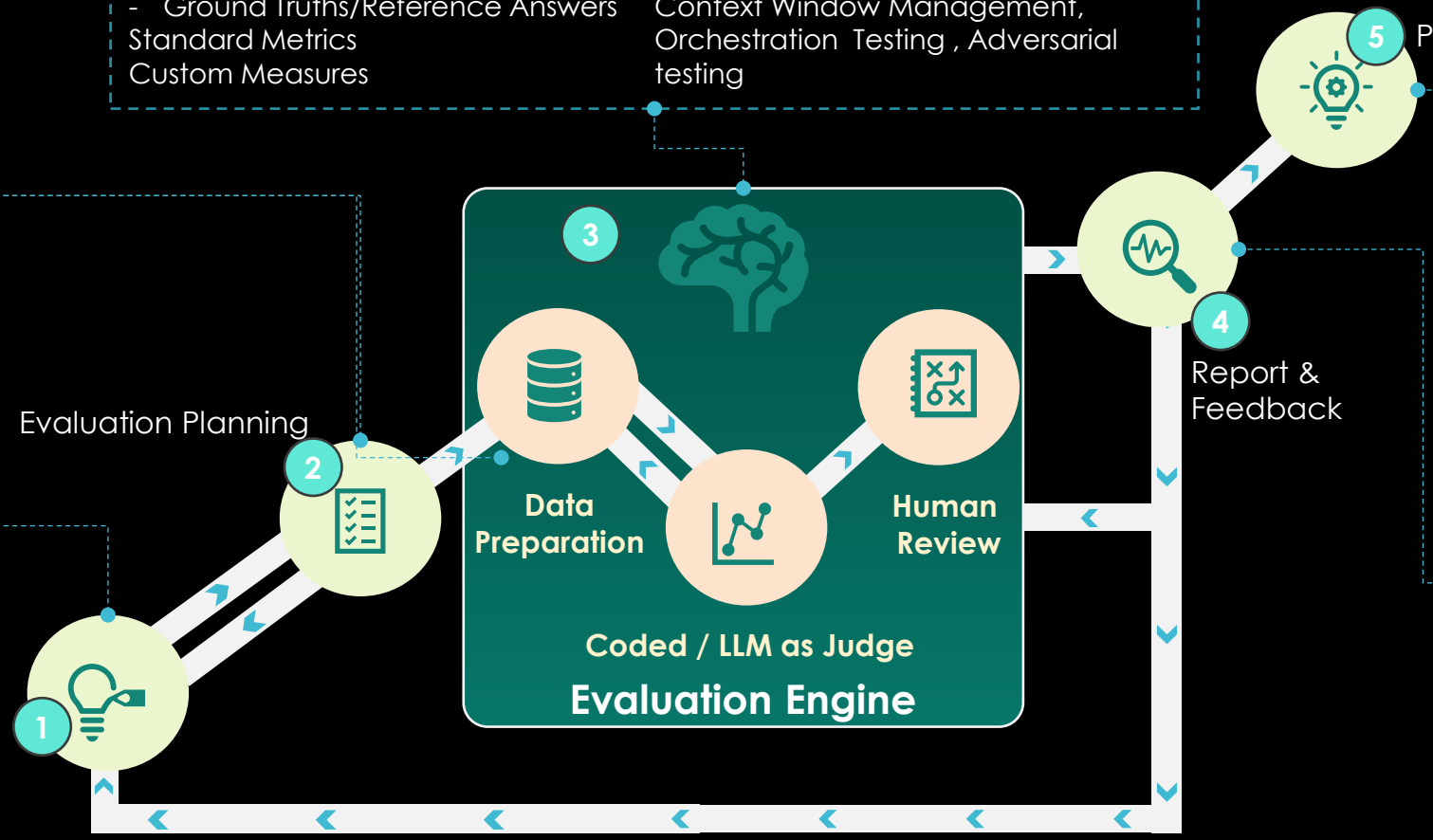
Standard Metrics
Custom Measures

NFR testing : Performance , Security
Grey box testing : Trace Validation , Tool Invocation , Mutational Testing, Context Window Management, Orchestration Testing , Adversarial testing

Coded(Quantitative)
LLM (Qualitative)
Dataset Requirements

Types of validations
- Response
- Retrieval
- Tools
- Cost
- Custom

AI System Schema:
- Architecture Docs
- Prompt Templates
- API Specs
- Sample Interactions



Continuous Monitoring
Dashboards
Drift check
Fairness, privacy and compliance

Evaluation Aggregator:
- Final Scores
- Failure Analysis
- Risk Insights
- Recommendations

AI Quality Report
- Accuracy
- Groundedness
- Tool Success
- Safety Score
- Insights & Risks

Completeness, Correctness, Relevance, And Adherence To Constraints, Reasoning, Planning Quality, Context Retention, Agent Trajectory, Decision-making, Memory Utilization, And RAG , LLM Usage, Data Storage, Cost Per Task, Response Time, Latency, Throughput, And Convergence For Optimal Performance, Robustness, Error Recovery, And Resilience , Ethical AI Practices, Privacy Safeguards, And Domain-specific Compliance

Agentic KYC System

Agentic STLC

Q & A

THANK YOU !!